STANDARD OPERATING PROCEDURE

ESP32 Energy Meter Gateway

Code identification, safe programming, low-voltage commissioning, and controlled 230 V AC work boundaries

Document control	Details
Document ID	EMS-FUSO-SOP-ESP32-001
Revision / issue date	Rev 1.0 / 14 October 2026
Applies to	The four tagged test-board files and the named EMS Fuso UDP receiver
Prepared from	Code review of tagged files, receiver/parser, and nearby project README files
Approval state	DRAFT FOR TECHNICAL REVIEW - NOT AUTHORIZED FOR LIVE WORK until site review and signatures

DANGER - 230 V AC CAN CAUSE FATAL ELECTRIC SHOCK, ARC FLASH, FIRE, OR EQUIPMENT DAMAGE. This is not a wiring diagram and does not authorize live work. Only appropriately licensed/qualified electrical personnel may design, connect, test, isolate, or energize mains circuits under the site permit and local electrical rules.

The firmware runs on low-voltage electronics; it does not make the mains measurement side safe. The reviewed source does not prove galvanic isolation, fuse coordination, enclosure rating, conductor sizing, protective earthing, creepage/clearance, or exact meter terminal connections.

1. Purpose, scope, and limitations

• This SOP identifies each supplied file and gives a controlled sequence for programming and commissioning the ESP32 Modbus gateway. It is based on code inspection, not physical inspection or electrical design approval.

• The integrated sketch reads one Modbus holding register and reports frequency. It currently polls one slave and one parameter; it is not a complete multi-meter energy-monitor implementation.

• Do not infer L/N, CT, voltage tap, fuse, terminal, or protective-earth wiring. Use approved drawings and the exact meter manual, reviewed by the responsible electrical engineer.

• Never use a breadboard, exposed jumper wires, or an unapproved USB-connected PC in an energized mains panel.

STOP WORK if the exact meter manual, approved schematic, isolation method, protective-device ratings, enclosure/earthing design, or safe isolation point is unknown or inconsistent.

2. Code file map - use one sketch at a time

File	Intended use	What it does / does not do
complete_testing.ino	Integrated gateway firmware; candidate for approved low-voltage bench testing.	ESP32 UART2 Modbus RTU read, DS3231 timestamp, SD CSV logging, W5500 Ethernet and UDP JSON transmission. Reads slave 1 / holding register 156 / two words as FLOAT32 every 5 seconds. Not multi-meter.
modbus_elmeasure.ino	Serial-only Modbus diagnostic sketch.	Reads register 156 from slave 1 over UART2 and prints raw Modbus frames/registers and decoded float to USB Serial. No RTC, SD, Ethernet, UDP or dashboard.
rtc_sdcard.ino	RTC and SD module bench-test sketch.	Checks DS3231 and SD; logs a test row to /modbus_log.csv every 5 seconds and prints file contents. No Modbus, meter, Ethernet or UDP.
app.py	Minimal UDP packet display tool for bench diagnostics only.	Binds all local interfaces on UDP 10011 and prints datagrams. Not the EMS Fuso dashboard/backend; does not validate or persist packets. Can conflict with another listener on the same host/port.

Arduino IDE rule: each .ino has its own setup() and loop(). Put the selected sketch alone in its own correctly named sketch folder before compiling. Do not combine the test sketches in one Arduino sketch folder; the IDE may concatenate .ino files and create duplicate entry points.

3. Integrated gateway reference (complete_testing.ino)

Subsystem	Code setting	Meaning / verification
ESP32 target	Assumes classic ESP32 with separate VSPI/HSPI.	Select exact board/module. Confirm boot-strapping and flash configuration. GPIO12 is W5500 MISO and a strapping pin on common classic ESP32 variants; verify reset-time level and board design.
RS485 / Modbus	UART2 RX=GPIO32, TX=GPIO33, DE/RE=GPIO4; 9600 baud, 8E1; slave 1.	Verify serial settings and A/B polarity from exact device documentation. Code does not prove transceiver galvanic isolation.
Meter data	Holding-register address 156; two 16-bit words; low-word-first FLOAT32 decode.	Confirm function/address base, word order, scaling and range with exact register map and known reference. Nearby project notes list different models/registers; do not copy blindly.
W5500	MISO=12, MOSI=13, SCK=14, CS=15, reset=25.	Global SPI. Verify wiring and startup on exact board/module.
Network	Device 192.168.0.229/24; gateway .1; DNS 8.8.8.8; destination 192.168.0.159:10011; local UDP 8888.	Confirm route, destination, unique IP/MAC and firewall/OT policy. Nearby README files show different gateway/destination values; code is not verified site truth.
Identity / interval	PLANT_ID=20246; DEVICE_ID=ESP32_20246; poll 5000 ms.	Confirm backend registration and identifiers. Only one slave/parameter is read.
DS3231 RTC	I2C SDA=21, SCL=22.	If power was lost, code sets compile-time date/time; compare/set to approved time. Compile time is not a live time source.
SD card	CS=27; SCK=18, MISO=19, MOSI=23; HSPI; /modbus_log.csv.	CSV includes timestamp, IDs, raw words, Frequency_Hz and status. Verify write and retrieval.

4. Source review findings - resolve before release

Priority	Finding	Required disposition
BLOCKER	UDP contract mismatch. Firmware sends one flat object (plant_id, device, slave_id, frequency_hz, etc.). EMS Fuso meter_service.process_reading() reads payload.device and payload.meters (array). It does not consume plant_id/frequency_hz. A packet can send successfully but produce no meter-data row.	Agree one schema with backend owner. Adapt firmware or parser to registered device identity and expected meter array/fields (meter ID, status, timestamp, parameter names). Prove persistence and dashboard visibility; Serial 'Telemetry sent' is not end-to-end success.
BLOCKER	Network settings conflict with nearby project documentation: sketch uses gateway .1 and destination .159; plant READMEs show gateway .254 and destination .41.	Network owner confirms subnet, gateway, receiver, UDP port, unique address/MAC and firewall; update approved constants only.
BLOCKER	Meter model/register assumptions are unverified. Sketch uses address 156 and low-word-first float; nearby notes identify other models/registers, including 158/2699.	Obtain exact meter manual; verify function, address base, 9600 8E1, slave ID, word/byte order, scaling, units and range against known reference.
HIGH	Electrical/logic isolation is not evidenced. A common MAX485 module may output 5 V logic to ESP32 (not 5 V tolerant); many modules are not galvanically isolated.	Use documented 3.3 V-compatible or properly level-shifted/isolated interface. Electrical engineer approves isolation, grounding, surge protection and separation.
HIGH	GPIO12 is W5500 MISO; on common classic ESP32 modules it is a boot strapping pin. Peripheral reset state can affect boot/flash configuration.	Validate exact board and MISO behavior during reset; revise pin design if necessary and test repeated cold boots.
HIGH	RTC lost-power recovery sets compile time, not actual time.	Set/check RTC against approved reference; record timezone and error. Do not rely on compile timestamp for audit-grade records.
MEDIUM	app.py is print-only UDP receiver on 10011 bound to 0.0.0.0. It is not the backend and conflicts if another receiver uses that host/port.	Use only isolated bench diagnostics. Stop it before real EMS Fuso listener. Restrict UDP to trusted OT network.
MEDIUM	Integrated sketch reports peripheral readiness but continues polling if some modules fail; Ethernet link is checked only during initialization.	Acceptance tests must reject missing required services and define/test behavior for link loss, RTC/SD failure and recovery.

5. Roles and prerequisites

Role	Responsibility
Qualified electrical person	Permit, LOTO, absence-of-voltage verification, mains design/connection, protection, earthing, enclosure inspection and energization approval.
Controls/firmware engineer	Board/core/libraries, pin/interface verification, Modbus map/schema, bench tests, backups and change record. No unauthorized live electrical work.
Network/backend owner	IP/MAC/UDP path, device registration, receiver process/port, packet schema, firewall/OT segmentation, proof of persisted readings.
Commissioning lead	Approved drawings, risk assessment, test instruments, checklists, stop-work decision and final sign-off.

Required documents and equipment

[ ] Approved drawings and exact meter, transceiver, ESP32/W5500/SD/RTC manufacturer documents.

[ ] Site permit/LOTO procedure, risk assessment, suitable PPE/tools, and calibrated/rated instruments for qualified personnel.

[ ] ESD-safe low-voltage bench, USB/programming PC, correct board/core/libraries, known-good isolated Modbus simulator or approved isolated source.

[ ] Network approval, server test access, SD/data handling procedure, code revision and firmware version record.

6. 230 V AC safety and stop-work rules

NO LIVE WORK. Before opening a panel, changing wiring, continuity testing, or USB programming near the gateway: isolate every relevant source, apply lockout/tagout, prove the tester, verify absence of voltage at the work point, then re-prove the tester. Follow site rules and local law.

[ ] Only a licensed/qualified electrical worker may handle 230 V AC conductors, meter voltage inputs, CT circuits, panel entry, fuses or energization.

[ ] Treat every conductor and terminal as live until isolated and verified. Account for alternate feeds, backfeed, stored energy, shared neutrals and induced voltage under site procedure.

[ ] Never connect mains voltage to ESP32 GPIO, USB, SD, RTC, W5500, RS485 logic pins, breadboards or low-voltage module terminals. Firmware pins are low-voltage logic only.

[ ] Keep mains and SELV/low-voltage wiring separated by an engineered enclosure/barrier, approved terminals, wiring methods, creepage/clearance and component ratings. Code alone provides no insulation barrier.

[ ] Use only meter/sensor, fuses, connectors, wiring and enclosure rated for voltage, fault current, environment, installation category and local code. A 230 V display value is not a component rating.

[ ] Do not connect a USB PC to circuitry that may be mains-referenced. Isolate and verify dead first. Any exceptional energized diagnostic method needs separately engineered isolation and site authorization; this SOP does not authorize it.

[ ] Use shock/arc PPE and barriers selected by site risk assessment. Keep the area dry and controlled, restrict access and maintain an emergency response plan.

[ ] Stop for damaged insulation, exposed live parts, unexpected voltage, uncertain labels/wiring, heat, smell, arcing, noise, trips or implausible readings. Do not bypass protection or re-energize to try again.

Emergency: do not touch a person/equipment still in contact with an energized source. Follow site emergency response; trained responders isolate power and summon emergency services as required.

7. Programming procedure (de-energized bench only)

[ ] Authorize the change. Record operator/date, asset ID, source filename/revision/hash, previous firmware, target board, test plan and recoverable backup.

[ ] Move equipment to a safe bench. Remove/lock out associated 230 V AC and meter voltage inputs; verify absence of voltage. Disconnect field conductors where the approved drawing requires. Do not program inside an energized panel.

[ ] Put the selected .ino alone in its own correctly named sketch folder. Keep complete_testing.ino, modbus_elmeasure.ino and rtc_sdcard.ino separate; do not combine sketches with duplicate setup()/loop().

[ ] Open in approved Arduino IDE/build system. Select exact ESP32 board/module, port and flash/partition settings from approved configuration. Do not guess board variant, flash mode or voltage.

[ ] Verify approved core/libraries: ModbusMaster; Ethernet/EthernetUDP and ArduinoJson (StaticJsonDocument API) for integrated code; RTClib; plus core SPI, SD, Wire, math/string support. Record versions; no ad-hoc production updates.

[ ] Review slave ID, function/register/address base, representation, serial format, pins, IP/gateway/destination, plant/device identity, UDP port, poll period and clock against approved documents. Never duplicate an assigned IP/MAC.

[ ] Compile/Verify before upload. Resolve errors. Review warnings about board/library/pins. A successful compile proves build syntax only, not safety, protocol correctness or dashboard ingestion.

[ ] With AC isolated and verified dead, connect USB, upload, wait for success, and use the approved board procedure. Do not hot-plug USB where it could energize exposed field circuitry.

[ ] Open Serial Monitor at 115200 baud. For integrated firmware, inspect RTC, SD, Ethernet, UDP and Modbus status. Record and resolve every FAILED/WARNING state; do not waive failures silently.

[ ] Remove USB before field reconnection unless an approved test method permits it. Keep mains isolated during Section 8 testing.

8. Low-voltage functional checks and acceptance

Perform with 230 V AC measurement circuits disconnected/isolated. Prefer a safe isolated Modbus simulator or manufacturer-approved isolated test fixture.

[ ] Cold-start repeatedly with W5500 connected as required; confirm boot and no GPIO12 strap/flash issue. Record board/module revision.

[ ] Modbus: confirm serial settings, slave ID, function, register/address base and two-word response against meter manual. Compare raw words and decoded float with known test value. Reject NaN, infinity, wrong units, impossible or unstable range.

[ ] RS485: verify transceiver logic voltage, DE/RE direction, A/B polarity, bus termination/bias and galvanic isolation against exact documentation and drawings.

[ ] RTC: compare timestamp with approved clock; record timezone/offset. Test missing RTC/lost-power behavior. Compile-time fallback is not suitable for traceability unless explicitly approved.

[ ] SD: verify card, header, append behavior, CSV fields/status, card removal procedure and data retrieval.

[ ] Ethernet: confirm W5500, link, approved static IP/gateway, unique IP/MAC and route.

[ ] UDP: capture datagram; confirm receiver/interface/port, exact schema, device registration, meter mapping and actual persisted/dashboard row. Sending or printing is not persistence.

[ ] Stop backup/testing_ems_board/app.py before starting EMS Fuso listener; both bind UDP 10011 on the same host.

[ ] Test Modbus timeout, invalid float, missing RTC/SD, cable/link loss and reboot recovery. Failure must not be mistaken for a fresh valid reading.

[ ] Save serial log, CSV excerpt, captured packet, database/dashboard evidence, software versions and test operator/date. Mark or remove test data per site policy.

9. Field installation and controlled energization

Only the authorized electrical person executes mains-side work. Firmware/controls personnel stay outside the panel unless qualified and authorized.

[ ] Verify approved drawings match physical meter, panel, protective devices, sensor/CT, conductor size and enclosure. Stop if any model, label or terminal differs.

[ ] Apply permit and LOTO to all relevant sources. Verify absence of voltage with site-approved prove-test-prove method at the point of work. Secure exposed parts.

[ ] Install measurement circuit, protection, earthing/bonding, isolation barrier, strain relief and enclosure only per exact approved drawings, manufacturer instructions and local rules. This SOP intentionally provides no mains-terminal wiring instructions.

[ ] With power isolated, qualified person inspects polarity, terminations/torque records, protection, earthing, covers, clearances, labels and separation against drawings.

[ ] Keep ESP32 field connection isolated during required pre-energization checks. Close/secure covers; clear tools/personnel; confirm work boundary.

[ ] Energize only under approved site procedure with the responsible electrical person in control. Do not probe exposed energized parts for this SOP.

[ ] Observe through protected/remote interfaces: startup status, known-range values, timestamp, SD logging and backend persistence. If absent, implausible or intermittent, isolate before inspection; no live wiring changes.

[ ] Record as-built differences and obtain electrical, firmware, network and commissioning sign-off before unattended operation.

10. Normal operation and shutdown

Normal operation

• At startup, confirm identity and no unresolved RTC, SD, W5500/link, UDP or Modbus failure. Keep gateway in approved enclosure with covers secured.

• Verify data freshness and plausible frequency against a known reference; prove dashboard persistence, not only serial output or UDP send.

• Review CSV/backend data under site retention policy. Protect SD/logs from unauthorized modification; maintain backups and clock practice.

• Do not change firmware, register maps, network settings or hardware without change control and repeat acceptance tests.

Shutdown / service

[ ] Stop dashboard/debug process gracefully; export required logs and record last known-good state.

[ ] For access or firmware change, isolate and LOTO all associated AC sources; verify absence of voltage before opening/touching field wiring.

[ ] Disconnect USB and field cables only in approved sequence. Protect SD card, configuration and source backup.

[ ] After service, repeat applicable low-voltage and electrical inspection/acceptance before authorized re-energization.

11. Troubleshooting - safe actions only

Symptom	Checks	Safe disposition
No boot/reboot	Exact board, USB source, reset log, GPIO12/W5500 reset, short circuits.	Disconnect field wiring and AC first. Pin changes require reviewed design; do not defeat protection.
Modbus timeout/value	Serial settings, ID, map/address base, transceiver voltage, A/B, DE/RE.	Never swap live. Isolate before rewiring; verify with simulator/manual.
RTC/SD failure	I2C pins 21/22, module compatibility, battery; SD format/capacity/CS/SPI/write errors.	No energized field rewiring. Preserve logs before formatting; do not accept invalid timestamps.
Ethernet/UDP absent	W5500/reset, duplicate address, subnet/gateway/destination, cable/switch/firewall, receiver port.	Use approved network plan and authorized packet capture; restrict to trusted OT network.
UDP sends/no dashboard row	Flat firmware JSON vs backend device + meters[] schema; registration and field names.	Release blocker. Correct under change control and prove persisted dashboard record.
Heat/odor/noise/arcing/trip	Potential electrical fault. Do not diagnose by touching/probing energized equipment.	Stop, keep clear and invoke site electrical emergency/isolation procedure.

12. Commissioning record and approvals

Complete before release. Attach drawing reference, exact equipment model/serial, firmware revision/hash, test evidence and risk assessment.

Record item	Entry
Site / panel / asset ID	_____
ESP32 board/module and revision	_____
Meter model / firmware / serial	_____
Approved drawing / revision	_____
Approved meter register/function/word order	_____
Network / receiver / schema version	_____
Firmware filename / source revision / hash	_____
Arduino core + library versions	_____
LOTO / absence-of-voltage record	_____
Low-voltage test evidence location	_____
Backend persisted-reading evidence	_____
Open deviations / corrective actions	_____
Electrical person name / signature / date	_____
Firmware/controls engineer / signature / date	_____
Network/backend owner / signature / date	_____
Commissioning lead approval / date	_____

Appendix A - packet contract to verify

Illustrative shape currently emitted by the integrated sketch; values are not approved configuration:

```
{"plant_id":"20246","device":"ESP32_20246","slave_id":1,"timestamp":"...","parameter":"Freq_Hz","register":156,"raw_word_1":123,"raw_word_2":456,"status":"OK","frequency_hz":50.0}
```

EMS Fuso services/meter_service.py reads payload.device and payload.meters as a list of meter objects with id/status and measurement fields (for example freq). The backend uses registered device identity and meter configuration. Firmware and server must be deliberately aligned; confirm the agreed final schema with the backend owner and prove persistence.

Appendix B - files reviewed

File	Review basis
backup/testing_ems_board/complete_testing.ino	Configuration; Modbus direction/conversion/read; RTC, SD, Ethernet, CSV, UDP, setup/loop.
backup/testing_ems_board/modbus_elmeasure.ino	ModbusDebugStream, register 156 test, raw frames, Serial output.
backup/testing_ems_board/rtc_sdcard.ino	DS3231/SD initialization, CSV logging/readback and loop.
backup/testing_ems_board/app.py	Socket bind on UDP 10011 and raw datagram print; no persistence/dashboard logic.
EMS Fuso services/meter_service.py and udp_server.py	Receiver schema, device registration, database persistence and UDP receive path.
EMS Fuso README and plant README files	Nearby conflicting network/register examples; not authoritative for this test board.

FINAL RELEASE GATE: This code-derived SOP is not an electrical design certificate. A qualified electrical engineer/site safety authority must review actual hardware, drawings, isolation/protection, 230 V installation method and local compliance before field installation or energization.